MT

MICROSCOPE TEAM

Microscope Team Bot

Feature Documentation

University Team | Telegram Bot CMS | Oct 2026

14	18	4	5x5
Feature Groups	Proposed	Roles	Pagination

Part 1: Current Bot Features

[Student] ?????? (Student Role)

Feature	Description
Browse Materials	Browse: Department -> Subject -> Category -> File, with 5x5 pagination at every level
Download Files	View and download PDF, video, audio, or direct links from Firebase
Smart Search	Search across all bot materials regardless of department
Subscribe to Subjects	Subscribe to a subject to receive notifications when new content is added
Study Tables	Register for group study schedules (Jadawel)
Contact Admin (Inbox)	Send direct messages to the admin from the bot
Membership Request	Submit a join request to the Contributors team, awaiting admin approval

[Contrib] ????????? (Contributor Role)

Feature	Description
Upload Files	Upload new content (PDF/video/audio/link) to any track, subject, category - 5x5 paginated path selection
Quick Add	Re-upload to the last used path in fewer steps
My Materials	View and delete files uploaded by this contributor, fully paginated
Edit Flow (RBAC)	Rename / Delete / Move any file AFTER verifying a time+use-limited auth code
My Stats	View total uploaded files and personal activity metrics
Auth Code Entry	Enter the temporary code issued by admin to unlock edit permissions; code is consumed on use

[Tools] ?????? (Admin Role)

Feature	Description
Full Dashboard	Central dashboard for all management operations
Manage Tracks & Subjects	Add / remove departments (Tracks) and subjects with 5x5 pagination
Manage Contributors	View contributor list, approve / reject join requests
Auth Code Generator	Generate a 6-digit random code configurable with: max uses + duration in minutes. Presets + custom settings. Auto-expires on either condition.
Revoke Code	Instantly revoke the current active code before it expires
Edit Files (Direct)	Rename / Delete / Move any file without needing a code
Broadcast	Send a message to all bot users
Bot Statistics	View total files, users, and contributors
RBAC System	Every action is protected by a role check (Student/Contributor/Admin/Super

Admin)

[Crown] ?????? ???? (Super Admin Role)

Feature	Description
All Admin Powers	Everything listed in the Admin section above
Ownership Transfer	Transfer Super Admin privileges to another user
Promote / Demote	Change any user's role between Admin <-> Contributor <-> Student

[PC] ?????? ????? ???? (Technical Features)

Feature	Description
Universal 5x5 Pagination	ALL menus across ALL roles show 5 items/page with Prev/Next buttons and X/Y page counter
Firebase Realtime DB	All content, users, and codes stored in Firebase RTDB
Callback Byte Safety	All Telegram callback data <= 64 bytes using Utils.shorten / Utils.lengthen
Missions System	Task assignment to HR committees displayed on the web portal
Web Portal (team.html)	Microscope Team showcase page with colors #102250/#1e40af/#00d2ff
HR Portal (hr.html)	Separate internal evaluation system for the team

Part 2: Proposed Features for Future Updates

[Books] 1. ?????? ????????? ????????? (Academic & Study Hub)

#	Feature	Description
1	Learning Paths (?????? ??????)	Step-by-step map guiding students on how to study each subject: which topic to start with, which book to use, and how to connect concepts.
2		
3	Review Schedule Generator (???? ????? ????????)	Auto-split the syllabus across days remaining before the exam. Student enters exam date + number of topics, bot generates the schedule.
4		
5	Academic Requirements Map (????? ?????????)	Visualize subject prerequisites to help students plan their semester course registration.

[Tools] 2. ????? ?????? ?????? (Student Tools & Services)

#	Feature	Description
1	PDF Maker (???? PDF ??????)	Convert photos of sheets/notes sent to the bot into a single downloadable PDF file.
2		
3	Programming Tools Library (????? ????? ???????)	Organized guide with links and instructions to install essential dev environments (VS Code, Python, Git, etc.) by subject/year.
4		
5	Internship & Courses Radar (????? ?????????)	Periodic notifications about summer internship opportunities, grants, and programming courses with direct application links.

[Game] 3. ??????? ???????? (Gamification & Engagement)

#	Feature	Description
1	Content Rating System (????? ???????)	Rating buttons ([OK] Helpful / [x] Needs improvement) below every file or explanation to surface quality content.
2		
3	Virtual Badges (??????? ?????????)	Award titles and icons to active users based on activity: 'Star of the Batch', 'Content Creator', 'Daily Challenger', etc.
4		
5	Easter Eggs (??????? ??????)	Hidden surprising responses that appear when certain words or phrases are typed accidentally, adding personality to the bot.

[Team] 4. ????? ??? ????????? ???? HR (Team Management)

#	Feature	Description
1	HR Portal & Team Management (????? ??? HR)	Secret section with special code for admins: track meeting attendance, rate member performance, manage committees and internal roles.
3	Auto Certificate Generator (?????? ????? ????????)	Issue personalized appreciation certificates for active members with their names and team signature as PDF, delivered directly via the bot.

Summary

Implemented & Live:	14 Feature Groups
Proposed for Future:	18 Features across 4 Categories
Tech Stack:	Dart 3.7.3 | Firebase RTDB | Telegram Bot API
Brand Colors:	#102250 (Navy) #1e40af (Royal Blue) #00d2ff (Cyan)
Auth Code System:	Time-bound + Use-limited | Auto-expires on either condition
Pagination:	Universal 5x5 across ALL roles & ALL menus
Last Updated:	October 2026